

Dynamic Memory Forest: Constructing and Tracing Conversational Trajectories for Long-Term Conversation

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Abstract

While large language models (LLMs) have made significant progress in expanding their context windows, they still face great challenges in effectively organizing and utilizing long-term memory to maintain conversation consistency and coherence. Summarizing historical conversations has achieved remarkable performance, which, however, loses conversational trajectory and association, making it difficult to precisely combine memories from different sessions in response to current queries. To address this, we propose the **Dynamic Memory Forest (DMF)**, a novel Consolidation-then-Growth framework for long-term open-domain conversation, which simulates the consolidation and growth processes of human memory by dynamically organizing long-term conversation histories into a memory forest of memory trees. To be specific, inspired by the principles of synaptic consolidation and plasticity from Cognitive Science, we first consolidate each session into memory units that preserve thematic coherence ("**Consolidation**"). Then, we first structure these units into memory trees and then grow the forest by dynamically connecting them through an evolutionary grafting mechanism, called *Group Relative Voting Optimization*, which mimics synaptic connection to decide whether a new memory tree should be grafted onto the existing forest or grow independently ("**Growth**"). For retrieval, we design an *Entropy-Driven Memory Walk*, constructing a logically coherent memory path via a navigation policy that prioritizes exploring high-entropy nodes. Experiments on three long-term conversation datasets show that our DMF significantly outperforms baselines in enhancing response generation for LLMs.

CCS Concepts

• **Information systems** → **Users and interactive retrieval.**

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1 Introduction

Recent advances in Large Language Models (LLMs) have been remarkable, particularly in expanding their context windows to accommodate millions of tokens [5, 8, 45]. This has significantly improved their performance in open-domain conversations, allowing for strong response generation based on extensive immediate context. Despite this progress, they still face great challenges in effectively organizing and utilizing long-term memory when dealing with discrete and complex conversation episodes [9, 21, 24, 27, 48, 49]. This suggests that efficient memory mechanisms are still needed to manage the conversation history and generate consistent and coherent responses.

The current paradigm focuses on summarizing historical conversations as memory units [1, 4, 18, 20, 26, 34, 47, 51, 55, 58] with remarkable results. However, they often fail to build meaningful associations between memories scattered across different sessions, resulting in a loss of conversational trajectory and association to precisely combine memories from isolated instances in response to current queries. This limitation prevents LLMs from developing a deep and evolving understanding of the conversation, keeping them from providing truly context-aware and human-like interactions. This points towards the need for a memory mechanism that, much like human memory, not only stores information but also actively consolidates and grows with new interactions.

According to Cognitive Science, human memory is not a static archive but a process of Consolidation and Growth [10, 28, 31–33]. New experiences are actively clustered and used to establish

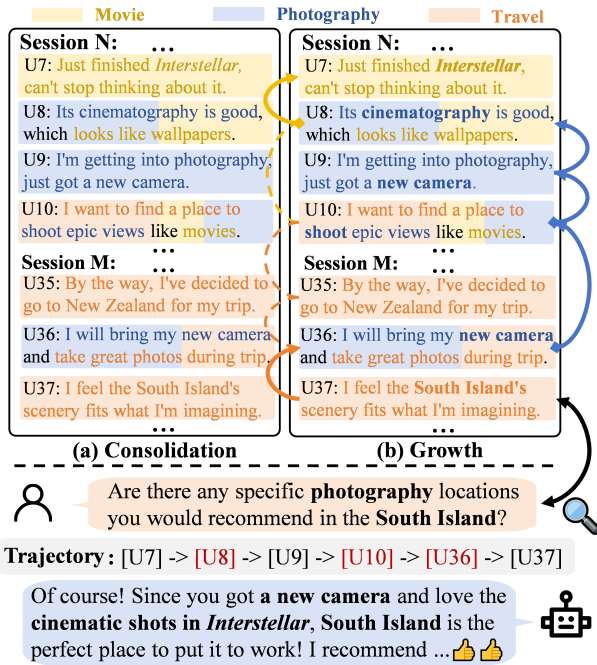


Figure 1: An illustration of memory consolidation, growth and walk of our DMF. Memories from conversations are first consolidated into thematic groups (a) and grow connections (b). For a given query, a coherent conversational trajectory is constructed by tracing connections within clusters and leaping between them via bridge nodes (eg., U8, U10, U36). This retrieved trajectory enables the generation of context-aware and logically consistent responses.

connections between memory nodes, causing the entire memory structure to grow constantly. As shown in Figure 1 (a), humans tend to organize memories into thematic groups, where each color block represents a distinct cluster. Furthermore, according to synaptic plasticity, the brain dynamically links related memories to form a complex and cohesive memory network. This is illustrated in Figure 1 (b), where memories from different clusters are connected to facilitate memory growth. This process often relies on bridge nodes that share attributes across multiple themes. Therefore, we argue that by first consolidating conversation history into thematic clusters and then growing connections between related memories, we can restore the lost conversational trajectory and achieve more effective memory utilization in long-term conversations.

To reach this goal, we propose the Dynamic Memory Forest (DMF), a novel Consolidation-then-Growth framework to generate better responses for long-term conversation. Specifically, for the **Memory Consolidation** stage, we get inspiration from synaptic consolidation, which encodes information and converts it into long-term form [11], and organize utterances from each session into clustered memory units, which create stable and coherent memory trees to preserve both thematic and temporal information. Further, inspired by synaptic plasticity that “Cells that fire together,

wire together” [14], for the **Memory Growth** stage, these trees are dynamically woven into an evolving memory forest. This is achieved by connecting new memory units to the existing structure through an evolutionary grafting mechanism, *Group Relative Voting Optimization* (GRVO), which employs a voting-based strategy to determine the optimal connection points between memory trees. For retrieval, we design the *Entropy-Driven Memory Walk* (EDMW). It simulates associative recall by starting from the most relevant memory to the query and performing an exploratory pathfinding via a navigation policy that prioritizes exploring high-entropy bridge nodes, ultimately providing LLMs with a logically coherent memory path. Experimental results on three long-term conversation datasets show that our DMF can enhance the ability of memory utilization for LLMs in long-term conversations, thereby leading to improved response generation. **The contribution of this work can be summarized as follows:**

- We explore a new paradigm to consolidate and grow memory for long-term open-domain conversation, which can effectively organize and utilize long-term memory.
- We are the first to dynamically grow a memory forest through a novel evolutionary grafting mechanism and then navigate it via an entropy-driven memory walk, which endows our framework with the ability to model conversational trajectory and association, thereby constructing a coherent memory path from discrete histories.
- Experimental results demonstrate the superiority of our proposed DMF framework over the prevailing summary-based paradigm across 128K and even 1M long-context LLMs in long-term response generation.

2 Related Work

Memory-Augmented Generation. Effectively modeling historical information as memory is a central challenge in achieving realistic, long-term open-domain conversations. The current approaches aim to build plug-and-play memory banks [4, 20, 26, 34, 47, 55, 58]. Within this trend, a popular paradigm is to compress entire sessions into condensed summaries and user-specific facts [4, 20, 58], or to recursively update a rolling memory summary [47]. Other research also incorporates temporal dynamics, emphasizing the impact of time intervals on memory relevance [34, 55]. **Different from these methods that rely on compressed summaries which often lose detail, our work proposes a new paradigm that restores the fine-grained conversational trajectory and association by organizing scattered histories into navigable structures.**

Structured Retrieval-Augmented Generation. Recent advancements have integrated structured knowledge into Retrieval-Augmented Generation (RAG) methods to enhance information organization and retrieval [2, 13, 15, 22, 44, 50, 54]. Approaches like MemWalker [3] and RAPTOR [41] construct hierarchical graph structures to enable recursive retrieval and summarization of texts. Similarly, GraphRAG [12] and MemTree [40] organize memory into knowledge graphs or dynamic trees to improve context-awareness and reasoning. **Different from these methods that often perform retrieval on isolated memory units, which leads to a loss of intrinsic associations between sessions, our method**

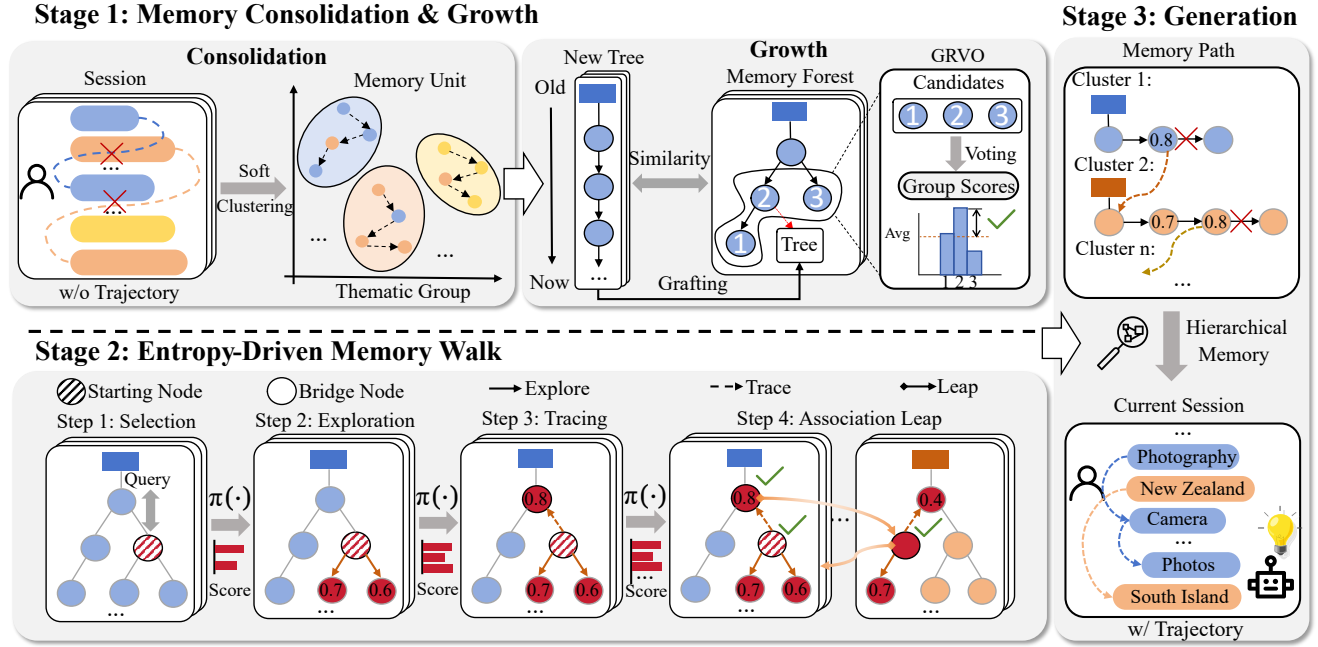


Figure 2: Illustration of our Consolidation-then-Growth memory utilization framework.

dynamically connects these discrete histories into a cohesive memory forest to restore the conversational trajectory.

Agentic Memory Management. Another trend employs autonomous agents to manage memory [6, 19, 25, 30, 36, 46, 52]. For instance, Chhikara et al. [6] utilize graph-based representations to capture complex relational structures. Xu et al. [52] link memories as structured notes that dynamically evolve through interconnected indexing. Moreover, Kang et al. [19] introduce an OS-inspired hierarchical storage system comprising short-, mid-, and long-term units. **Different from these management strategies where agents manually determine memory updates, our work proposes a bio-inspired paradigm where the memory forest grows organically through an evolutionary grafting mechanism, ensuring robust and coherent connections.**

Cognitive Memory Modeling. Cognitive Science posits that human memory involves first consolidating new experiences [10, 28, 31] into thematic clusters and then dynamically linking them, akin to synaptic plasticity [14], to form a complex and evolving memory network [32, 33]. In contrast, prevailing long-term memory mechanisms often focus on summarization, which lose conversational trajectory and associations, while agentic memory management and structured RAG treats memory as a static archive for one-off, similarity-based retrieval, failing to capture the evolving connections between memories across sessions. **Inspired by Cognitive Science, our DMF builds upon these cognitive principles, realizing memory consolidation by clustering conversations into thematic groups, and growth by structuring them into memory trees and connecting them through an evolutionary**

grafting mechanism that mimics synaptic plasticity, thereby building a cohesive and evolving memory forest.

3 Methodology

As shown in Figure 2, the core idea of DMF is to dynamically construct long-term conversation histories into a memory forest composed of memory trees. By simulating the processes of memory consolidation and growth, our DMF enables efficient organization and precise recall of complex conversational histories, leading to context-aware response generation. This section details our DMF and all source code will be made publicly available upon publication.

3.1 Memory Consolidation

The goal of memory consolidation is to transform the unstructured stream of dialogue from a single session into structured memory units with thematic cohesion.

Soft Clustering. After a session concludes, we first process all user utterances $u \in \mathcal{S}_{session}$. We begin by reducing the dimensionality of the utterance embeddings $e(u)$ using Uniform Manifold Approximation and Projection [29]. Subsequently, we apply a Gaussian Mixture Model (GMM) [39] to the lower-dimensional embeddings for soft clustering, where the optimal number of clusters $\mathcal{K}$ is automatically determined by the Bayesian Information Criterion [42]. The GMM yields a posterior probability $\gamma(u, k) = p(C_k | e(u))$ for each utterance’s membership in cluster C_k . Based on this, we calculate the thematic entropy for each utterance:

$$H(u) = - \sum_{k=1}^{\mathcal{K}} \gamma(u, k) \log \gamma(u, k) \quad (1)$$

This entropy value quantifies the utterance’s thematic uncertainty and is the core basis for memory association.

Memory Unit. Each cluster C_k formed through this process is treated as an independent memory unit U_k . We assign all utterances with a posterior probability $\gamma(u, k)$ exceeding a certain threshold δ to this unit:

$$U_k = \{u \in \mathcal{S}_{\text{session}} \mid \gamma(u, k) > \delta\} \quad (2)$$

Critically, this process allows us to identify bridge nodes: utterances with high thematic entropy $H(u)$, as these nodes have significant membership in multiple thematic units, they serve as natural semantic hubs connecting different memory trees. This design breaks down complex conversation histories and consolidates related conversation flows, laying the foundation for subsequent structured organization and dynamic growth.

3.2 Memory Growth

The value of memory lies not only in storage but also in its ability to dynamically grow and form connections. This section describes how memory units are organized into tree structures and evolve into a memory forest.

Memory Tree. A simple set of utterances loses the inherent temporal information of a conversation. To solve this, we convert each unordered memory unit U_k into an ordered memory tree T_k . Specifically, we first sort the utterances within the unit by their original timestamps t_u to form an ordered dialogue chain $L_k = (u_1, u_2, \dots, u_m)$, where $t_{u_i} < t_{u_{i+1}}$. We then utilize LLMs to create the unit’s core summary as a root node r_k and link it to the head of the chain u_1 . This gives each memory unit a tree structure $T_k = (r_k, L_k)$ that preserves local temporal logic.

Memory Forest and GRVO. The evolution of long-term memory often involves revisiting, deepening, or extending past conversations. Therefore, we design a dynamic growth mechanism that allows a new memory tree T_{new} to form a meaningful grafting with the historical memory forest $F = \{T_1, T_2, \dots, T_N\}$. Specifically, we first calculate the cosine similarity between the root node $r_{\text{new}} \in T_{\text{new}}$ and all summary nodes in forest F to obtain the most similar historical subtree. We then form the candidate grafting node set C from all utterance nodes within this single most relevant tree. Inspired by [43], we introduce GRVO, which is driven by a similarity-based voting mechanism and an advantage score to ensure that the grafting between the forest F and the new memory trees is of high quality and significance. Specifically, each utterance u_j in T_{new} casts its cosine similarity score as a vote for its most similar candidate grafting node. To select the optimal grafting node, GRVO calculates an advantage score for each candidate $c_i \in C$:

$$A(c_i) = S(c_i) - \frac{1}{|C|} \sum_{c_k \in C} S(c_k) \quad (3)$$

Here, $S(c_i)$ is the total cosine similarity-based vote score received by the candidate c_i , and $|C|$ is the number of candidates in the set C . A high positive advantage score indicates that a node is an exceptional grafting point, significantly outperforming the average. If the score of the best candidate $A(c_{\text{best}})$ exceeds a predefined threshold τ , we graft T_{new} to c_{best} to complete the dynamic growth.

Otherwise, to prevent incorrect grafting, T_{new} is added to the forest as a standalone tree. Compared with selecting absolutely similar scores, this design can more accurately determine whether to graft or keep a tree independent, ensuring that the nodes in each tree are highly related.

3.3 Entropy-Driven Memory Walk

Traditional retrieval is a one-off fetch, whereas human recall is a dynamic process of association. To this end, we model this with a memory retrieval algorithm, called EDMW, which generates an optimal memory path $M^* = (n_1, n_2, \dots, n_L)$ for each query q .

Specifically, we first select the Top- K memory trees on the cosine similarity between the current query q and their root summaries. Then, we find the most query-similar utterance node within each of those trees. From n_{current} , the walk considers downward exploration, upward tracing, and associative leap, with a key constraint: once a downward path is explored, the upward trace cannot cross into sibling sub-trees. The associative leap is the core of EDMW, which allows the walking path to jump to the bridge node in other memory trees when it encounters a bridge node, thus realizing cross-tree memory association. At each step, our method needs to decide on the next move based on q , the current node, and the bridge nodes in the forest. To select the best path among numerous possibilities, we design a navigation policy $\pi(n_{\text{next}} | n_{\text{current}}, q)$ to score each potential n_{next} :

$$\pi(n_{\text{next}} | n_{\text{current}}, q) = \underbrace{\text{sim}(n_{\text{next}}, q)}_{\text{Global Relevance}} \cdot \underbrace{(1 + \text{sim}(n_{\text{next}}, n_{\text{current}}))}_{\text{Coherence Amplifier}} \cdot \underbrace{(1 + \text{Norm}(H(u_{\text{next}})))}_{\text{Entropy Amplifier}} \quad (4)$$

where u_{next} is the utterance corresponding to node n_{next} , $H(u_{\text{next}})$ is its pre-computed thematic entropy, $\text{sim}(\cdot, \cdot)$ is the cosine similarity, and $\text{Norm}(\cdot)$ is a normalization function. This policy ensures that EDMW stays closely aligned with the query while maintaining conversation coherence, and prioritizes the exploration of high-entropy bridge nodes with richer information. The walk terminates once the selected downward path to a leaf node or upward path to the root of current subtree has been fully explored.

3.4 Response Generation

Memory Path. Upon completion of our EDMW, the final memory path M^* is meticulously constructed for LLMs. Specifically, all utterance nodes visited along the optimal path are collected and deduplicated. This unique set of nodes is then sorted according to their global timestamps to form a temporally coherent narrative.

Hierarchical Memory. To further enrich this narrative with hierarchical memory, we inject the summary from a memory tree’s root node directly before the first utterance of its corresponding thematic group appears in the sorted path. The final response r^* is defined as follows:

$$r^* = \underset{r \in \mathcal{R}}{\operatorname{argmax}} P_{\text{LLM}}(r | q, M^*) \quad (5)$$

where $\mathcal{R}$ represents a set of all possible responses.

Table 1: Automatic evaluation (%) of generation performance (average of sessions). “Bold Font” means the highest results, while “Underlined Font” means second-highest results. *B-4 = BLEU-4, R-L = ROUGE-L, and Bert = BertScore.

Backbone	Methods	CC				MSC				GC			
		B-4	R-L	Bert	Mauve	B-4	R-L	Bert	Mauve	B-4	R-L	Bert	Mauve
GPT-4o	Full History (128K)	<u>1.79</u>	<u>17.41</u>	47.79	<u>55.73</u>	<u>1.21</u>	<u>15.12</u>	49.17	<u>54.36</u>	0.66	11.43	<u>36.57</u>	25.12
	<i>Structured Retrieval-Augmented Generation</i>												
	MemWalker (2023a)	0.89	13.25	45.32	42.17	0.76	11.58	46.85	43.21	0.54	9.87	33.21	21.45
	RAPTOR (2024)	0.95	14.12	46.15	43.88	0.82	12.33	47.23	44.57	0.58	10.12	34.15	22.89
	GraphRAG (2024)	1.05	14.89	46.98	45.23	0.91	12.87	48.05	46.12	0.63	10.55	35.43	24.32
	MemTree (2025)	1.02	14.45	<u>47.83</u>	44.56	0.85	12.54	<u>49.20</u>	45.33	0.59	10.29	34.78	23.56
	<i>Agentic Memory Management</i>												
	Mem0 (2025)	1.02	14.57	45.85	46.92	0.69	12.78	45.68	45.61	0.53	9.42	34.08	23.43
	A-Mem (2025)	1.21	15.12	46.01	50.04	0.88	13.07	46.79	50.74	0.61	10.14	35.48	26.23
	MemoryOS (2025)	1.14	15.46	45.73	47.87	0.97	13.86	47.55	46.28	0.69	10.33	35.87	24.58
	<i>Memory-Augmented Generation</i>												
	MemoChat (2023)	0.72	12.56	45.78	35.61	0.83	12.63	47.93	53.21	0.74	10.94	35.03	22.51
	MemoryBank (2024)	1.08	15.14	47.27	45.95	1.03	13.74	48.39	45.51	0.64	10.05	35.78	23.32
	Rsum (2025a)	1.01	14.57	47.12	46.51	0.97	13.99	48.43	51.99	<u>1.06</u>	16.16	35.77	27.48
	COMEDY (2025)	0.67	11.31	46.18	39.51	0.61	11.07	47.19	48.86	0.51	9.91	34.01	24.81
	LD-Agent (2025b)	1.37	15.78	46.42	50.16	1.02	14.05	47.76	48.63	0.72	10.47	35.96	25.94
	THEANINE (2025)	1.27	14.84	45.69	54.23	0.94	13.55	47.42	53.64	0.79	10.23	35.77	28.97
	DMF (Ours)	2.98	20.46	49.71	72.55	1.41	16.22	49.37	55.08	1.23	<u>15.43</u>	41.19	30.12
Gemini2.5	Full History (1M)	1.57	17.51	47.51	72.04	0.89	13.61	47.59	55.61	0.78	10.05	35.76	25.44
	<i>Structured Retrieval-Augmented Generation</i>												
	MemWalker (2023a)	0.58	6.33	35.53	32.74	0.52	5.89	34.12	31.25	0.43	5.12	30.15	22.89
	RAPTOR (2024)	0.76	9.91	41.59	35.97	0.68	8.76	40.23	34.88	0.55	8.12	32.89	24.56
	GraphRAG (2024)	0.88	10.45	42.15	36.82	0.79	9.56	41.33	35.74	0.62	9.15	33.91	25.67
	MemTree (2025)	0.95	10.75	42.33	36.24	0.84	9.92	41.87	35.91	0.67	9.45	34.56	26.12
	<i>Agentic Memory Management</i>												
	Mem0 (2025)	1.09	15.88	44.73	52.94	0.93	12.92	45.88	49.46	0.66	9.64	35.67	26.93
	A-Mem (2025)	1.18	14.97	45.38	51.86	0.75	12.42	46.85	49.71	0.76	10.28	36.12	25.94
	MemoryOS (2025)	1.26	15.75	45.97	51.93	0.91	12.78	45.76	48.94	0.71	9.53	35.58	26.47
	<i>Memory-Augmented Generation</i>												
	MemoChat (2023)	1.21	13.71	44.91	51.01	0.83	11.77	45.56	<u>56.81</u>	0.57	10.13	34.11	28.58
	MemoryBank (2024)	<u>1.96</u>	<u>17.64</u>	45.91	57.64	<u>1.05</u>	<u>14.54</u>	49.83	51.69	0.88	10.89	34.74	26.86
	Rsum (2025a)	1.56	16.97	<u>48.17</u>	63.41	1.09	14.43	47.67	52.24	0.61	10.55	35.27	27.15
	COMEDY (2025)	1.55	16.63	46.71	57.01	0.93	11.89	46.73	55.39	0.67	10.22	33.57	25.09
	LD-Agent (2025b)	1.43	16.17	45.78	60.42	0.96	12.54	45.47	49.93	0.74	9.83	35.49	27.96
	THEANINE (2025)	1.64	17.02	45.23	<u>75.42</u>	1.07	14.27	46.01	55.64	<u>0.91</u>	<u>11.45</u>	<u>36.98</u>	<u>30.29</u>
	DMF (Ours)	3.67	21.14	48.81	77.23	1.49	15.46	<u>47.96</u>	66.81	1.34	11.95	39.54	37.69

4 Experiments

4.1 Experimental Settings

Datasets. Following [34, 55], we evaluate our method on three long-term open-domain conversation datasets: **Conversation Chronicles** (CC) [17], **Multi-Session Chat** (MSC) [51], and **GapChat** (GC) [55]. These datasets are human-verified and built through a meticulous crowdsourcing pipeline, purpose-built for the simulation and evaluation of long-term, context-dependent conversations. Consistent with the settings in [34], we randomly sampled 50 episodes from each test set, yielding a total of 250 sessions for each generation experiment.

Model and Baselines. We evaluate on two closed-source long-context LLMs: 1) **GPT-4o (128K)** [16], the gpt-4o-2024-11-20 version. 2) **Gemini2.5 (1M)** [7], the gemini-2.5-pro-preview-03-25 version. For comprehensive evaluation, we compare our DMF against various methods: 1) **Full History**: which directly feeds the entire conversation history into the model without filtering. 2) **Four structured RAG methods**: **MemWalker** [3], **RAPTOR** [41], **GraphRAG** [12], and **MemTree** [40], which organize external information into hierarchical trees or graphs to enhance retrieval precision. 3) **Three agentic memory management methods**: **MemoryOS** [19], **Mem0** [6], and **A-Mem** [52], which employ

autonomous agents to dynamically store, update, and retrieve user-specific memories. 4) **Six memory-augmented generation methods: MemoChat** [26], **MemoryBank** [58], **COMEDY** [4], **Rsum** [47], **LD-Agent** [20], and **THEANINE** [34], which utilize summarization or compression techniques to distill critical information from long contexts. 5) **Two search strategies: Greedy Search** and **Beam Search**, which are employed to traverse the search space for optimal memory selection.

Evaluation Metrics. We comprehensively evaluate our DMF on three types of metrics. 1) **Automatic Metrics.** Following [34], we use BLEU-4 [37], ROUGE-L [23], Mauve [38], and BertScore [56] to automatically evaluate response generation. 2) **G-Eval Metrics.** Following [17, 53], we introduce GPT4o-as-a-Judge [34, 57] to evaluate Gemini2.5’s response generation on five dimensions: *Engagingness*, which measures the ability to capture user interest; *Humanness*, which evaluates the naturalness and empathy of the response; *Coherence*, which assesses the logical flow and clarity; *Consistency*, which checks for factual alignment with history; and *Memorability*, which gauges the effective utilization of past details. Each metric is scored on a scale of 1-5, with 1 being the worst and 5 being the best. Normalisation is taken in G-Eval experimental results to maintain a better visualisation. 3) **Human Metrics.** Following [34], humans evaluate the winning performance of different methods on response generation and memory retrieval. Following [17], we employ 5 in-house evaluators who assess 50 randomly selected Gemini2.5’s generations and memorizes across datasets.

Implementation Details. Following [34], we also employ *text-embedding-3-small* [35] for similarity calculations and set LLMs’ temperature to 0.75. To ensure fair comparisons, all baselines are implemented using their original prompts. For our hyperparameter, we set $\delta = 0.3$, $\tau = 50\%$, and $K = 3$. **Given that Gemini2.5 demonstrated the best overall performance in the automatic evaluations, we select it for all subsequent experiments.**

4.2 Main Results

DMF improves response generation by preserving conversational trajectory and association. Our proposed DMF significantly improves response generation by preserving conversational trajectory and association, a key limitation in summary-based paradigms that organize memories into isolated units. This advantage is demonstrated across three long-context LLMs (GPT-4o and Gemini2.5), where DMF consistently outperforms all baseline methods in automatic evaluations (Table 1), with particularly notable gains in Mauve. This is further validated by the G-Eval results (Figure 4), where DMF excels in *Coherence* and *Consistency*, and confirmed by human evaluation (Figure 3). The strong human preference for DMF’s generation quality is a direct consequence of providing a coherent memory path instead of disconnected summaries, leading to more logical responses.

A long context window does not equal strong memorability. Table 1 also reveals an important insight: simply having a long context window is not enough for long-term conversations. We compare our DMF against Full History, where LLMs have access to the entire conversation within its context. In all metrics, our method achieves much higher scores than Full History. It suggests

Baselines:	Generation					Memory				
	20%	40%	60%	80%	100%	20%	40%	60%	80%	100%
Full History	76%	10%	14%			N/A				
MemoChat	80%	10%	10%			78%	14%	8%		
MemoryBank	74%	16%	10%			76%	16%	8%		
Rsum	78%	14%	8%			80%	14%	6%		
COMEDY	70%	20%	10%			72%	16%	12%		
LD-Agent	72%	14%	4%			70%	10%	20%		
THEANINE	68%	18%	4%			70%	14%	6%		

Legends: ■ Ours wins ■ Tie ■ Baseline wins

Figure 3: Human evaluation of generation and memory compared to memory-augmented generation baselines.

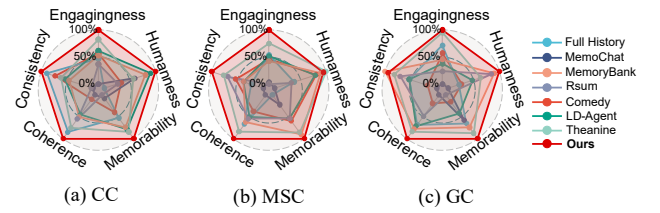


Figure 4: G-eval of Gemini2.5’s response generation compared to various baselines.

that a well-structured memory mechanism like our method is more effective than relying on a long but unstructured context window. Moreover, different LLMs also have different memorability.

Better memory coherence leads to better generation. Our method achieves better results in human preference evaluations because it produces more coherent and logical responses. This is supported by multiple metrics. First, the high Mauve scores in Table 1 indicate that the text generated by DMF is stylistically closer to human writing. Second, the G-Eval results in Figure 4 show the comprehensive superiority of our method across five dimensions, especially the higher scores achieved by our method in terms of *Coherence* and *Consistency* in the GPT-4o evaluation. Finally, and most importantly, the human evaluation in Figure 3 confirms this. The high win-rate for DMF in Memory is directly linked to its high win-rate in Generation. This suggests that because our method can build a more coherent memory path, it leads to generations that humans strongly prefer.

4.3 Ablation Study

To validate the effectiveness of each component within our DMF, we conduct a comprehensive ablation study. As Table 2 shows, removing any single component leads to a degradation in performance, which confirms that all parts of our design contribute to the final result:

Effectiveness of Consolidation-then-Growth Paradigm. As shown in Table 2, removing the Consolidation stage (w/o Consolidation) or the Growth stage (w/o Growth) both results in a noticeable drop in performance across all datasets. This confirms that our

Table 2: Ablation study on Gemini2.5, where w/o means without. * MP = Memory Path, HM = Hierarchical Memory, CA = Coherence Amplifier, and EA = Entropy Amplifier.

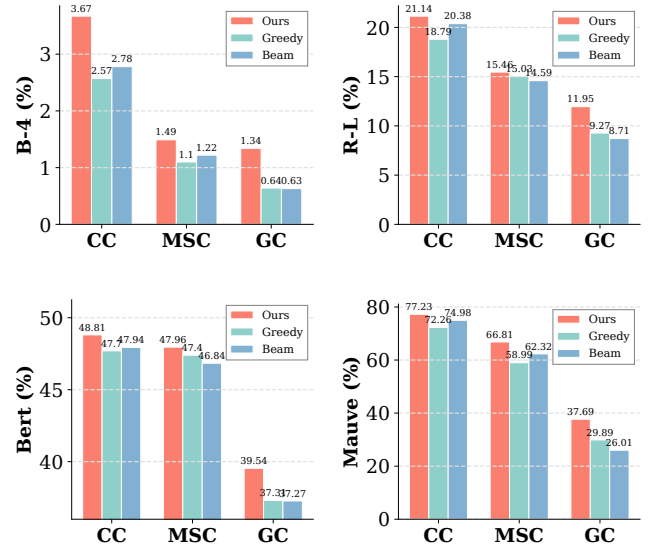
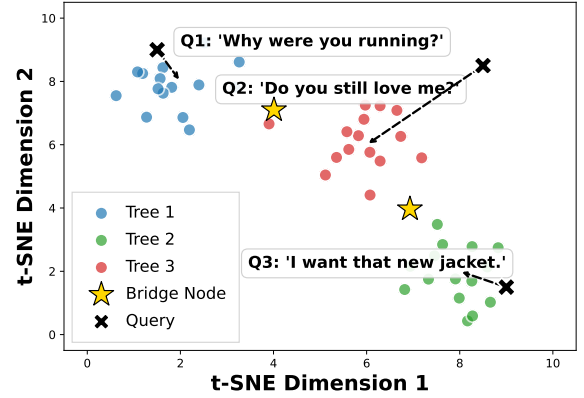
Datasets	Methods	B-4	R-L	Bert	Mauve
CC	DMF (Ours)	3.67	21.14	48.80	77.23
	w/o Consolidation	3.53	21.09	48.30	70.01
	w/o Growth	3.62	21.01	48.40	76.74
	w/o GRVO	3.25	20.53	48.76	71.51
	w/o EDMW	2.28	19.34	48.07	77.22
	w/o CA	3.45	20.75	48.33	76.31
	w/o EA	3.04	20.54	48.68	75.54
	w/o MP & HM	3.13	20.66	48.60	76.31
MSC	DMF (Ours)	1.49	15.46	47.96	66.81
	w/o Consolidation	1.32	15.12	47.51	56.57
	w/o Growth	1.29	15.09	47.63	59.33
	w/o GRVO	1.16	14.79	47.02	52.41
	w/o EDMW	1.15	14.92	46.85	53.31
	w/o CA	1.24	15.01	47.44	58.57
	w/o EA	1.23	14.88	47.29	58.83
	w/o MP & HM	1.31	15.17	47.28	63.62
GC	DMF (Ours)	1.34	11.95	39.54	37.69
	w/o Consolidation	0.80	10.38	36.38	34.83
	w/o Growth	0.89	10.56	37.22	32.67
	w/o GRVO	0.89	10.91	37.32	31.21
	w/o EDMW	0.79	10.04	37.38	29.79
	w/o CA	1.06	11.33	37.80	32.09
	w/o EA	0.98	10.50	37.20	30.44
	w/o MP & HM	0.81	10.18	36.34	28.33

paradigm of first structuring memories and then dynamically connecting them is fundamental to our DMF success.

Effectiveness of GRVO and EDMW. The removal of GRVO (w/o GRVO) for memory growth, or the EDMW (w/o EDMW) for memory retrieval, causes a significant decline in all metrics. The performance drop is particularly severe when removing EDMW. These results show that both GRVO and EDMW are highly effective and essential to the DMF.

Analysis of EDMW Navigation Policy π . Removing the Coherence Amplifier (w/o CA) or the Entropy Amplifier (w/o EA) both weakens the results. This indicates that guiding the memory walk to maintain local coherence and to prioritize high-entropy bridge nodes are both crucial strategies for constructing the optimal memory path.

Impact of Generation Formatting. Removing the structured Memory Path and Hierarchical Memory (w/o MP & HM) also leads to a significant performance drop. This shows that how the retrieved memories are organized and presented to LLMs is also a critical factor in generating high-quality final responses.

**Figure 5: Performance comparison with Greedy and Beam search baselines on CC, MSC, and GC datasets.****Figure 6: t-SNE visualization of a partial memory forest corresponding to a single episode. Different colors represent different memory trees.**

4.4 Visualization of Memory Distribution

Figure 6 visualizes the semantic distribution of memory nodes using t-SNE. In the plot, the dots represent memory nodes, and we can clearly see three distinct groups of colors. This indicates that our model successfully consolidates similar conversations into separate, well-organized memory trees. The black "X" marks represent new user queries. For instance, Query 1 ("Why were you running?") lands directly next to the blue tree, which shows that the model correctly maps the question to the relevant memory topic. Additionally, the yellow stars represent bridge nodes. These nodes sit in key semantic positions, acting as hubs to link different conversation threads. Overall, this demonstrates that our system effectively turns messy conversation history into clear, searchable clusters.

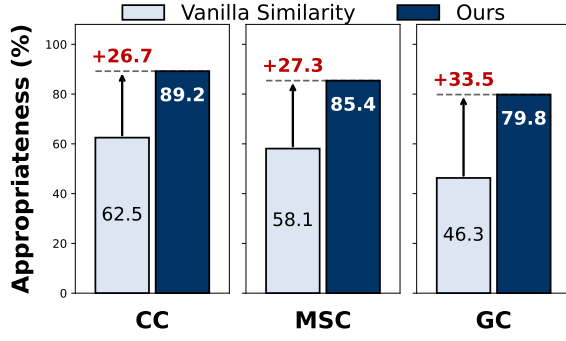


Figure 7: Evaluation of Grafting Appropriateness.

Table 3: Retrieval performance comparison.

Methods	Recall		NDCG	
	@5	@10	@5	@10
THEANINE (Memory-Augmented)	35.24	48.15	41.32	45.67
MemTree (Structured RAG)	32.18	44.50	43.89	46.12
MemoryOS (Agentic Memory)	38.56	52.33	45.21	48.95
DMF (Ours)	44.82	61.05	50.14	54.28

4.5 Comparison with Search Baselines

As illustrated in Figure 5, our method consistently outperforms classic Greedy and Beam search baselines across all metrics on the CC, MSC, and GC datasets. While traditional search strategies focus strictly on immediate relevance, often limiting themselves to local optima, our EDMW effectively utilizes high-entropy bridge nodes to perform associative leaps. This capability allows the model to construct a globally coherent memory path rather than merely retrieving surface-level matches, ultimately yielding responses that are logically superior and more human-like.

4.6 Analysis of Grafting Appropriateness

To evaluate the quality of memory connections objectively, we employ GPT-4o as a judge. We randomly sample 100 grafting decisions from each dataset. For each sample, GPT-4o review the new memory and the selected historical tree to determine if the connection was logical and appropriate. As shown in Figure 7, GRVO consistently outperforms the baseline, achieving high appropriateness scores (>85%) on continuous datasets like CC and MSC by effectively filtering semantic noise. The advantage is most profound on the sparse GC dataset, where GRVO maintains a high score of 79.8% compared to the baseline’s 46.3%. This substantial gap confirms that our group voting mechanism is essential for identifying logical bridges in disjointed, long-term conversations where simple similarity matching fails.

4.7 Retrieval Accuracy Analysis

To strictly evaluate the retrieval capability of our entropy-driven mechanism, we conduct a standalone retrieval experiment on 100 sessions randomly sampled from three datasets. Since open-domain

conversation datasets lack explicit retrieval labels, we utilize GPT-4o as a standard annotator to identify historical utterances essential for generating the ground-truth response. As shown in Table 3, our DMF significantly outperforms all baselines, achieving the highest Recall@10 of 61.05%. While methods like Theanine and MemoryOS struggle with long-range dependencies, our approach effectively leverages bridge nodes to traverse semantic clusters. This confirms that our framework can accurately retrieve logically relevant memories that are topologically distant but semantically crucial, rather than just relying on recency or surface-level similarity.

4.8 Case Study

Figure 8 illustrates a specific case demonstrating how our DMF framework connects discrete sessions to generate context-aware responses. In this scenario, the memory forest contains two distinct trees: a “Baking” tree and a “Fitness” tree. The “Baking” tree records that the user bought a new oven (A1) but found previous cakes too sweet (A2), while a grafted branch (C1-C2) reflects a new desire for healthy recipes. The “Fitness” tree records the user’s goal to lose weight (B1) and a “no sugar” constraint from their coach (B2). When the user asks, “What can I bake with my new oven?”, the model first identifies the starting node A1 in the Baking tree. It then traverses to the high-entropy bridge node C2 (“want healthy food”). This triggers an associative leap to the Fitness tree, connecting to node B2 (“no sugar”) due to the semantic bridge between “healthy” and dietary constraints. The path then traces upward to the root node B1 (“lose weight”), where the walk terminates to ensure the retrieval of the tree’s core theme. By sequencing retrieved nodes by timestamp into a memory path, the model synthesizes the “new oven” context with dietary constraints to logically recommend “sugar-free cookies”. This proves our method effectively recalls and combines information across different conversational episodes.

4.9 Efficiency Analysis

Token Consumption. As shown in Figure 9 (a), our method achieves the lowest total consumption. In contrast, MemTree consumes the most 4× ours in the final session—due to expensive tree updates after every sentence, followed by THEANINE and MemoryOS which also perform frequent updates. Our method minimizes maintenance costs via a batch processing strategy that consolidates memories only at the end of a session. Furthermore, retrieving concise summaries instead of raw texts keeps inference prompts short, achieving an optimal balance between performance and cost.

Time Consumption. Figure 9 (b) demonstrates our superior time efficiency. For construction, MemTree incurs the highest latency (over 20× ours) due to frequent per-turn reconstruction, whereas our session-level batch processing significantly reduces overhead. Regarding retrieval, while THEANINE suffers from complex graph traversal delays, our method maintains better efficiency, ensuring low latency suitable for real-time long-term interaction.

4.10 Hyperparameter Sensitivity Analysis

We investigate the impact of three key hyperparameters on model performance: the consolidation threshold (δ), the growth threshold (τ), and the number of retrieved trees (K), as shown in Figure 10.

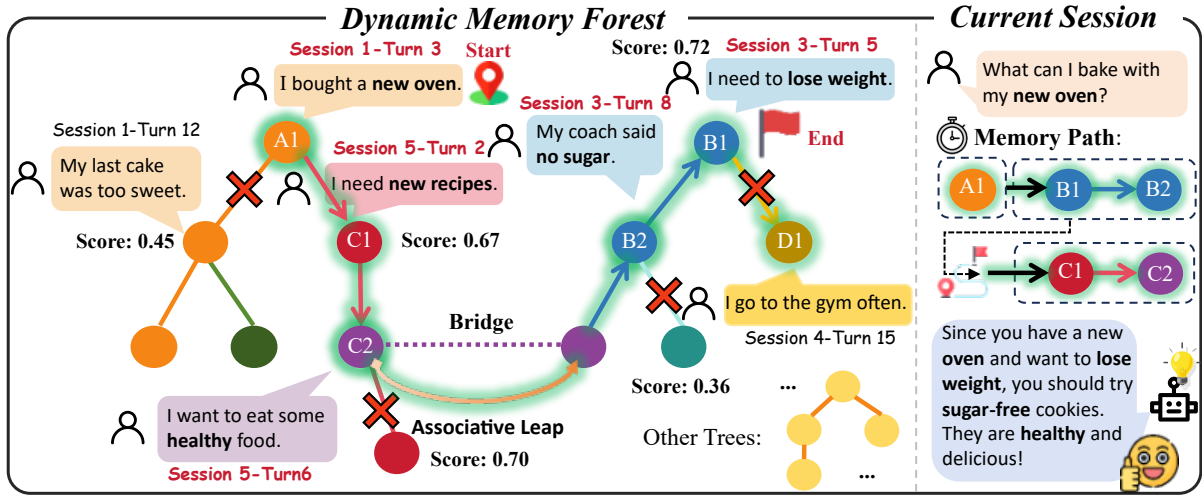


Figure 8: Case study illustrates how our DMF performs an associative leap from the "Baking" tree to the "Fitness" tree to generate a coherent response. * Note that the dialogues in the nodes are simplified and contain only key information for clarity.

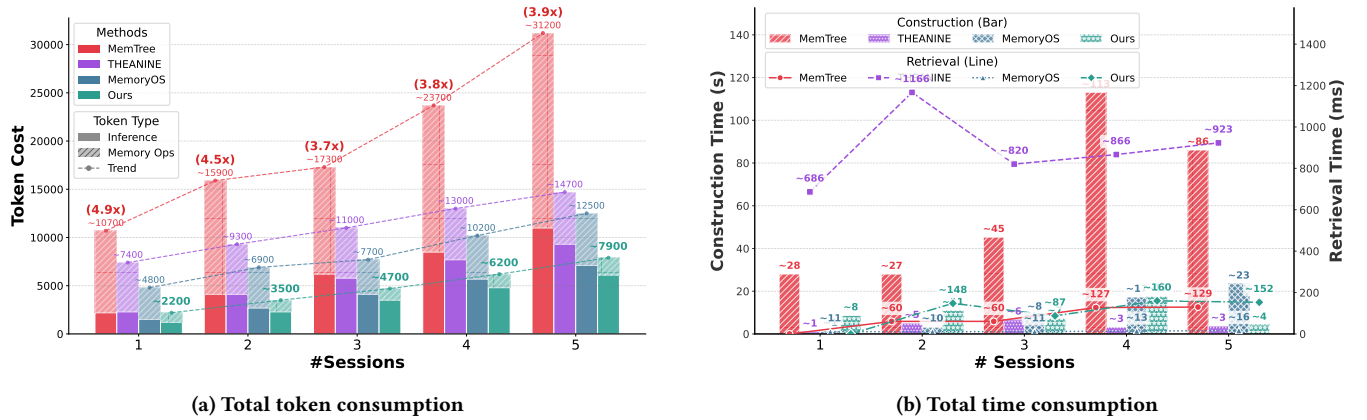


Figure 9: Efficiency analysis. (a) shows the total token consumption and (b) displays the total time consumption across sessions.

First, for the consolidation threshold δ , performance remains relatively stable across a wide range, peaking around $\delta = 0.5$. This suggests our soft clustering is robust, though extreme values slightly degrade coherence by either over-segmenting or over-merging topics. Second, the growth threshold τ exhibits an optimal point at 50%. Lower thresholds introduce noise by grafting irrelevant trees, while higher thresholds hinder memory growth by rejecting valid connections, leading to fragmentation. Finally, for retrieval size K , we observe a "sweet spot" at $K = 3$ to 5. Increasing K beyond this point introduces irrelevant context that distracts the generator, causing a slight decline in metrics like BLEU-4 and ROUGE-L. Overall, our method demonstrates strong robustness, consistently maintaining high performance without requiring exhaustive tuning.

5 Discussion

Table 4 presents the topological statistics of our constructed Memory Forest, validating the effectiveness of the Consolidation-then-Growth paradigm. Notably, the average tree depth (12.1) consistently exceeds the average number of nodes per memory unit (8.8), mathematically confirming that our evolutionary grafting mechanism successfully extends memory vertically rather than creating isolated clusters. Furthermore, the high graph reachability (74.4%) combined with a low bridge node ratio (14.8%) demonstrates that our Growth mechanism successfully connects disparate memory clusters into a coherent navigable structure, avoiding the fragmentation issues common in session-based approaches.

6 Conclusions

In this work, we propose DMF, a Consolidation-then-Growth framework that restores conversational trajectories for long-term open-domain conversation. By simulating the cognitive processes

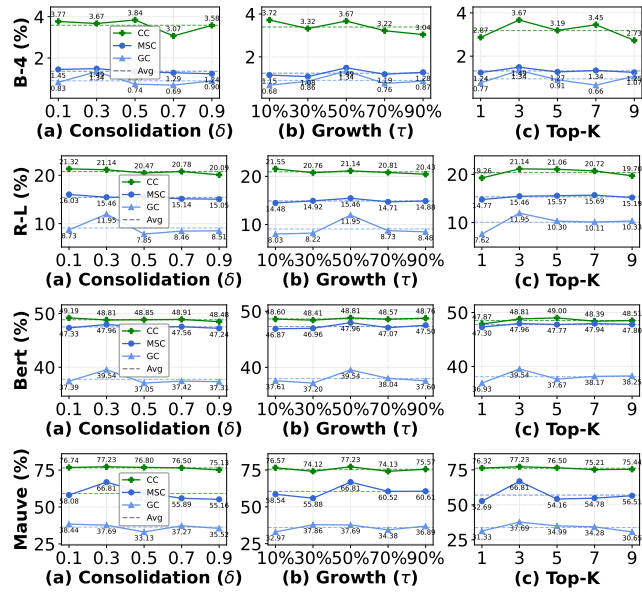


Figure 10: Parameter sensitivity analysis on four metrics: BLEU-4, ROUGE-L, BertScore, and Mauve. The dashed lines represent the average performance.

Table 4: Statistics of the constructed memory forest across three datasets. Bridge Ratio indicates the percentage of nodes functioning as semantic bridges. Reachability measures the average percentage of the forest accessible from a single root via bridge nodes.

Metric	CC	MSC	GC	Avg.
Avg. # of Trees	12.4	10.6	11.2	11.4
Avg. Tree Depth	12.2	11.5	12.7	12.1
Avg. Nodes per Unit	8.2	7.4	10.8	8.8
Bridge Node Ratio (%)	22.8%	11.2%	10.5%	14.8%
Graph Reachability (%)	71.1%	71.6%	80.5%	74.4%

of memory consolidation and growth, DMF transforms fragmented conversational history into a structured, navigable forest. Extensive experiments show that DMF significantly outperforms various baselines. Ultimately, our findings reveal that a well-structured memory mechanism is more critical than simply expanding the context window for generating consistent and human-like responses.

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